

Assignment 1 — Full Worked Solutions

MA1110: Calculus I, IIT Hyderabad

(Written for exam presentation; every proof/argument here is what you would actually write.)

Q1. Convergence of sequences

(i) $x_n = \frac{(-1)^n n}{n^2 + 1}$

Solution. For all n , $0 \leq |x_n| = \frac{n}{n^2 + 1} \leq \frac{n}{n^2} = \frac{1}{n} \rightarrow 0$. By the Sandwich Theorem, $|x_n| \rightarrow 0$, and hence $x_n \rightarrow 0$. ■

$$\boxed{x_n \rightarrow 0}$$

(ii) $x_n = \frac{3n^2 + \sin n - 4}{2n^2 + 3}$

Solution. Divide numerator and denominator by n^2 :

$$x_n = \frac{3 + \frac{\sin n}{n^2} - \frac{4}{n^2}}{2 + \frac{3}{n^2}}.$$

Since $\left| \frac{\sin n}{n^2} \right| \leq \frac{1}{n^2} \rightarrow 0$ and $\frac{4}{n^2}, \frac{3}{n^2} \rightarrow 0$, the algebra of limits gives

$$x_n \rightarrow \frac{3 + 0 - 0}{2 + 0} = \frac{3}{2}.$$

■

$$\boxed{x_n \rightarrow \frac{3}{2}}$$

(iii) $x_n = \frac{n+1}{\sqrt{n}}$

Solution. $x_n = \sqrt{n} + \frac{1}{\sqrt{n}} \geq \sqrt{n}$ for all n . Given any $M > 0$, choose $N > M^2$; then for $n > N$, $x_n \geq \sqrt{n} > M$. Hence $x_n \rightarrow \infty$: the sequence *diverges* (properly diverges to $+\infty$), it does not converge. ■

$$\boxed{\text{diverges } (x_n \rightarrow \infty)}$$

(iv) $x_n = \left(1 + \frac{1}{n^2}\right)^{n^2}$

Solution. Recall the standard fact $b_m := \left(1 + \frac{1}{m}\right)^m \rightarrow e$ as $m \rightarrow \infty$ (through integers). Here $x_n = b_{n^2}$, i.e. $\{x_n\}$ is the subsequence of $\{b_m\}$ obtained by taking $m = n^2$. Since $n^2 \rightarrow \infty$ as $n \rightarrow \infty$, and $b_m \rightarrow e$, every subsequence of $\{b_m\}$ also converges to e ; in particular $x_n = b_{n^2} \rightarrow e$. ■

$$\boxed{x_n \rightarrow e}$$

$$(v) \ x_n = \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2}{n^2}\right) \left(1 + \frac{3}{n^2}\right) \right]^{n^2}$$

Solution. Take logarithms:

$$\ln x_n = n^2 \sum_{i=1}^3 \ln \left(1 + \frac{i}{n^2}\right).$$

For fixed i , using $\ln(1+t) = t - \frac{t^2}{2} + O(t^3)$ with $t = i/n^2 \rightarrow 0$,

$$n^2 \ln \left(1 + \frac{i}{n^2}\right) = n^2 \left(\frac{i}{n^2} - \frac{i^2}{2n^4} + O(n^{-6}) \right) = i - \frac{i^2}{2n^2} + O(n^{-4}) \rightarrow i.$$

Summing over $i = 1, 2, 3$:

$$\ln x_n \rightarrow 1 + 2 + 3 = 6.$$

Since $t \mapsto e^t$ is continuous, $x_n = e^{\ln x_n} \rightarrow e^6$. ■

$$\boxed{x_n \rightarrow e^6}$$

Q2. Monotonicity

(Assumed reading, matching the linear/linear style of part (ii): $a_n = \frac{3n}{3n+1}$.)

$$(i) \ a_n = \frac{3n}{3n+1}$$

Solution. Write $a_n = 1 - \frac{1}{3n+1}$. Since $3n+1$ is strictly increasing in n , $\frac{1}{3n+1}$ is strictly decreasing, so $a_n = 1 - \frac{1}{3n+1}$ is **strictly increasing**. ■

strictly increasing

$$(ii) \ a_n = \frac{5n-2}{4n+1}$$

Solution. Compute $a_{n+1} - a_n$ by putting over a common denominator:

$$a_{n+1} - a_n = \frac{5n+3}{4n+5} - \frac{5n-2}{4n+1} = \frac{(5n+3)(4n+1) - (5n-2)(4n+5)}{(4n+5)(4n+1)}.$$

Numerator: $(20n^2 + 17n + 3) - (20n^2 + 17n - 10) = 13$. So

$$a_{n+1} - a_n = \frac{13}{(4n+5)(4n+1)} > 0 \quad \text{for all } n \geq 1.$$

Hence $\{a_n\}$ is **strictly increasing**. ■

strictly increasing

(iii) $a_n = \cos \frac{n\pi}{3}$

Solution. This sequence is periodic with period 6, cycling through

$$a_1 = \frac{1}{2}, \quad a_2 = -\frac{1}{2}, \quad a_3 = -1, \quad a_4 = -\frac{1}{2}, \quad a_5 = \frac{1}{2}, \quad a_6 = 1, \quad a_7 = \frac{1}{2}, \dots$$

Consecutive differences alternate in sign, e.g.

$$a_2 - a_1 = -1 < 0 \quad (\text{decrease}), \quad a_4 - a_3 = \frac{1}{2} > 0 \quad (\text{increase}).$$

Since $\{a_n\}$ has both a decrease and an increase between consecutive terms — and this pattern repeats forever, because the sequence is periodic with period 6 — it is neither eventually increasing nor eventually decreasing. Hence $\{a_n\}$ is **neither increasing nor decreasing**. ■

neither (oscillates, periodic)

Q3. Divergent sequences with convergent sum and product

Solution. Take

$$x_n = (-1)^n, \quad y_n = (-1)^{n+1} = -(-1)^n.$$

Both diverge. Consider $x_n = (-1)^n$. Its even subsequence is $x_{2k} = 1$ for all k , so $x_{2k} \rightarrow 1$; its odd subsequence is $x_{2k-1} = -1$ for all k , so $x_{2k-1} \rightarrow -1$. If (x_n) converged to some limit x , then by Theorem 15 (every subsequence of a convergent sequence converges to the same limit) both subsequences would have to converge to x ; but they converge to different values $1 \neq -1$, a contradiction. So (x_n) diverges (this is exactly the argument used in Example 13/40 of the notes). The identical argument applied to $y_n = -x_n$ shows (y_n) diverges too.

Sum converges:

$$x_n + y_n = (-1)^n + (-1)^{n+1} = (-1)^n - (-1)^n = 0 \quad \text{for every } n,$$

which is the constant sequence 0, so by Example 9, $x_n + y_n \rightarrow 0$.

Product converges:

$$x_n y_n = (-1)^n \cdot (-1)^{n+1} = (-1)^{2n+1} = -1 \quad \text{for every } n,$$

the constant sequence -1 , so again by Example 9, $x_n y_n \rightarrow -1$. ■

$$x_n = (-1)^n, \quad y_n = (-1)^{n+1}$$

Q4. If $x_n > 0$ and $\lim x_n^{1/n} = l > 0$, show $\lim [(n+1)x_{n+1}]^{1/n} = l$

Solution. Since $n+1 > 0$ and $x_{n+1} > 0$, we may split the n -th root of the product:

$$[(n+1)x_{n+1}]^{1/n} = (n+1)^{1/n} \cdot x_{n+1}^{1/n}.$$

Step 1: $(n+1)^{1/n} \rightarrow 1$. For $n \geq 2$, $n \leq n+1 \leq n^2$. Taking n -th roots (order-preserving for positive numbers),

$$n^{1/n} \leq (n+1)^{1/n} \leq (n^2)^{1/n} = (n^{1/n})^2.$$

By Exercise 13(1), $n^{1/n} \rightarrow 1$; hence $(n^{1/n})^2 \rightarrow 1$ too (Theorem 9, product rule). By the Sandwich Theorem (Theorem 12), $(n+1)^{1/n} \rightarrow 1$.

Step 2: $x_{n+1}^{1/n} \rightarrow l$. Let $y_m := x_m^{1/m}$; by hypothesis $y_m \rightarrow l$. The shifted sequence y_{n+1} (indices $n+1 \rightarrow \infty$ as $n \rightarrow \infty$) also converges to l , by Theorem 15.

Since $\frac{n+1}{n} = 1 + \frac{1}{n}$,

$$x_{n+1}^{1/n} = \left(x_{n+1}^{1/(n+1)}\right)^{(n+1)/n} = y_{n+1} \cdot y_{n+1}^{1/n}.$$

We show $y_{n+1}^{1/n} \rightarrow 1$. Since $y_{n+1} \rightarrow l > 0$, taking $\varepsilon = l/2$ in the definition of convergence gives N_0 with

$$\frac{l}{2} < y_{n+1} < \frac{3l}{2} \quad \text{for all } n \geq N_0.$$

Taking n -th roots (order-preserving for positive bases),

$$\left(\frac{l}{2}\right)^{1/n} \leq y_{n+1}^{1/n} \leq \left(\frac{3l}{2}\right)^{1/n} \quad \text{for } n \geq N_0.$$

By Example 25 (for any fixed $a > 0$, $a^{1/n} \rightarrow 1$), both outer bounds $\rightarrow 1$, so by the Sandwich Theorem, $y_{n+1}^{1/n} \rightarrow 1$.

By Theorem 9 (product rule), $x_{n+1}^{1/n} = y_{n+1} \cdot y_{n+1}^{1/n} \rightarrow l \cdot 1 = l$.

Conclusion. Combining Steps 1 and 2 via Theorem 9 (product rule):

$$[(n+1)x_{n+1}]^{1/n} = (n+1)^{1/n} \cdot x_{n+1}^{1/n} \rightarrow 1 \cdot l = l.$$

■

Q5. Convergence of $x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$

Solution. **Monotone increasing:** $x_{n+1} - x_n = \frac{1}{(n+1)!} > 0$, so $\{x_n\}$ is strictly increasing.

Bounded above: For $k \geq 1$, $k! = 1 \cdot 2 \cdot 3 \cdots k \geq 1 \cdot 2 \cdot 2 \cdots 2 = 2^{k-1}$ (the first factor is $1 \leq 2^0$, each of the remaining $k-1$ factors is ≥ 2), so $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$. Hence

$$x_n = 1 + \sum_{k=1}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \frac{1}{2^{k-1}} = 1 + 2 \left(1 - \frac{1}{2^n}\right) < 1 + 2 = 3.$$

So $\{x_n\}$ is bounded above by 3.

Conclusion: By the Monotone Convergence Theorem, $\{x_n\}$ converges. Its limit is, by definition,

$$\lim_{n \rightarrow \infty} x_n = \sum_{k=0}^{\infty} \frac{1}{k!} = e.$$

■

$\{x_n\}$ converges, and $x_n \rightarrow e$

Q6. The sequence $1, \frac{1}{2}, 1, \frac{2}{3}, 1, \frac{3}{4}, \dots$ converges to 1

Solution. The odd-indexed terms are $a_{2k-1} = 1$ for every k , and the even-indexed terms are $a_{2k} = \frac{k}{k+1}$.

The odd subsequence is constantly 1, so it converges to 1.

The even subsequence satisfies $a_{2k} = \frac{k}{k+1} = 1 - \frac{1}{k+1} \rightarrow 1$ as $k \rightarrow \infty$.

Now fix $\varepsilon > 0$. Since $a_{2k} \rightarrow 1$, there is K such that $|a_{2k} - 1| < \varepsilon$ for all $k \geq K$; take $N = 2K$. For any $n \geq N$: if n is odd, $a_n = 1$, so $|a_n - 1| = 0 < \varepsilon$; if n is even, $n = 2k$ with $k \geq K$, so $|a_n - 1| < \varepsilon$. Hence $|a_n - 1| < \varepsilon$ for all $n \geq N$, i.e. $a_n \rightarrow 1$. ■

$$\boxed{a_n \rightarrow 1}$$

Q7. $x_{n+1} = \frac{1}{2}(x_n + y_n)$, $y_{n+1} = \sqrt{x_n y_n}$, $x_1, y_1 > 0$: common limit

(This is the classical arithmetic–geometric mean (AGM) recursion; double-check against your copy that the printed recursion for y_{n+1} is $\sqrt{x_n y_n}$ and not $\sqrt{x_n + y_n}$ — the AGM version below is the standard textbook exercise that matches “converge to a common limit.”)

Solution. **Step 1: All terms are positive.** By induction: $x_1, y_1 > 0$, and if $x_n, y_n > 0$ then $x_{n+1} = \frac{1}{2}(x_n + y_n) > 0$ and $y_{n+1} = \sqrt{x_n y_n} > 0$.

Step 2: $x_n \geq y_n$ for $n \geq 2$ (AM $\geq$ GM). For any $a, b > 0$, $(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a + b \geq 2\sqrt{ab}$, i.e. $\frac{1}{2}(a + b) \geq \sqrt{ab}$. With $a = x_n$, $b = y_n$:

$$x_{n+1} = \frac{1}{2}(x_n + y_n) \geq \sqrt{x_n y_n} = y_{n+1} \quad \text{for all } n \geq 1,$$

so $x_n \geq y_n$ for every $n \geq 2$.

Step 3: $\{x_n\}_{n \geq 2}$ is decreasing, bounded below. For $n \geq 2$, using $x_n \geq y_n$ from Step 2,

$$x_{n+1} = \frac{x_n + y_n}{2} \leq \frac{x_n + x_n}{2} = x_n.$$

So $\{x_n\}_{n \geq 2}$ is decreasing. It is bounded below by y_2 : this follows once Step 4 shows $\{y_n\}$ is increasing, since then $x_n \geq y_n \geq y_2$ for all $n \geq 2$. By the Monotone Convergence Theorem (Exercise 11: a decreasing sequence bounded below converges), $x_n \rightarrow L$ for some L .

Step 4: $\{y_n\}_{n \geq 2}$ is increasing, bounded above. Since $x_n \geq y_n$,

$$y_{n+1} = \sqrt{x_n y_n} \geq \sqrt{y_n \cdot y_n} = y_n,$$

so $\{y_n\}_{n \geq 2}$ is increasing, and bounded above by x_2 (as $y_n \leq x_n \leq x_2$ for $n \geq 2$, using Step 3). By the Monotone Convergence Theorem (Theorem 11), $y_n \rightarrow M$ for some M .

Step 5: $L = M$. Taking $n \rightarrow \infty$ in $x_{n+1} = \frac{1}{2}(x_n + y_n)$ and using Theorem 9 (Algebra of Convergent Sequences),

$$L = \frac{1}{2}(L + M) \implies 2L = L + M \implies L = M.$$

■

$$\boxed{x_n, y_n \rightarrow \text{a common limit } L \text{ (the AGM of } x_1, y_1)}$$

Q8. Rational sequence $\rightarrow$ irrational limit; irrational sequence $\rightarrow$ rational limit

Solution. **(a) Rationals converging to an irrational.** For each $n \in \mathbb{N}$, apply Theorem 4 (Density of $\mathbb{Q}$ in $\mathbb{R}$) to the interval $(\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n})$ (a genuine interval, since $\frac{1}{n} > 0$): there exists a rational number x_n with

$$\sqrt{2} - \frac{1}{n} < x_n < \sqrt{2} + \frac{1}{n}, \quad \text{i.e.} \quad |x_n - \sqrt{2}| < \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$ (Example 10), the Sandwich Theorem gives $x_n \rightarrow \sqrt{2}$. Each x_n is rational by construction, and $\sqrt{2}$ is irrational.

(b) Irrationals converging to a rational. Let $y_n = \frac{\sqrt{2}}{n}$. Each y_n is irrational: if y_n were rational, then $\sqrt{2} = n y_n$ would be a product of two rational numbers, hence rational — a contradiction. By Theorem 9 (Algebra of Convergent Sequences) applied to $y_n = \sqrt{2} \cdot \frac{1}{n}$ with $\frac{1}{n} \rightarrow 0$ (Example 10), $y_n \rightarrow \sqrt{2} \cdot 0 = 0$, which is rational. ■

Q9. $x_1 = 1, x_{n+1} = \frac{1}{x_n + 2}$: convergence and limit

Solution. This recursion is not itself monotonic (it alternates above/below its limit), so we apply the Monotone Convergence Theorem to the even- and odd-indexed subsequences separately, then combine them via Exercise 15.

Step 1: The second-iterate map. Since $x_{n+2} = \frac{1}{x_{n+1} + 2} = \frac{1}{\frac{1}{x_n + 2} + 2}$, simplifying the compound fraction gives

$$x_{n+2} = \frac{x_n + 2}{2x_n + 5} =: g(x_n).$$

So both the odd-indexed terms $a_k := x_{2k-1}$ and the even-indexed terms $b_k := x_{2k}$ satisfy the *same* first-order recursion, $a_{k+1} = g(a_k)$ and $b_{k+1} = g(b_k)$, where $g(x) = \frac{x+2}{2x+5}$.

Step 2: Properties of g . For $x > 0$,

$$g(x) - x = \frac{(x+2) - x(2x+5)}{2x+5} = \frac{-2x^2 - 4x + 2}{2x+5} = \frac{-2(x^2 + 2x - 1)}{2x+5}.$$

The equation $x^2 + 2x - 1 = 0$ has roots $-1 \pm \sqrt{2}$; write $L = \sqrt{2} - 1 > 0$ for the positive root. Since the other root $-1 - \sqrt{2}$ is negative, for $x > 0$ we have $x - (-1 - \sqrt{2}) = x + 1 + \sqrt{2} > 0$ always, so $x^2 + 2x - 1 = (x - L)(x + 1 + \sqrt{2})$ has the same sign as $(x - L)$. Hence

$$g(x) - x \text{ has the opposite sign to } (x - L), \quad \text{for every } x > 0. \quad (*)$$

Also $g'(x) = \frac{1}{(2x+5)^2} > 0$ everywhere, so g is strictly increasing on $(0, \infty)$.

Step 3: The odd subsequence $a_k = x_{2k-1}$ decreases to L . We have $a_1 = x_1 = 1 > L$ (numerically $L = \sqrt{2} - 1 \approx 0.414$). If $a_k > L$, then since g is increasing and $g(L) = L$ (as L solves $x^2 + 2x - 1 = 0$), $a_{k+1} = g(a_k) > g(L) = L$; by induction $a_k > L$ for every k . By (*), since $a_k > L$ we get $a_{k+1} - a_k = g(a_k) - a_k < 0$, so $\{a_k\}$ is strictly decreasing. Being decreasing and bounded below (by L), it converges by the Monotone Convergence Theorem (Exercise 11); say $a_k \rightarrow L_1$.

Step 4: The even subsequence $b_k = x_{2k}$ increases to L . We have $b_1 = x_2 = \frac{1}{3} < L$. If $b_k < L$, then $b_{k+1} = g(b_k) < g(L) = L$ (as g is increasing), so by induction $b_k < L$ for every k . By

(*), $b_{k+1} - b_k = g(b_k) - b_k > 0$, so $\{b_k\}$ is strictly increasing, bounded above by L ; it converges by Theorem 11 (Monotone Convergence Theorem); say $b_k \rightarrow L_2$.

Step 5: Identify the limits. Taking $k \rightarrow \infty$ in $a_{k+1} = g(a_k) = \frac{a_k + 2}{2a_k + 5}$ and using Theorem 9 (Algebra of Convergent Sequences — valid since $2L_1 + 5 \neq 0$):

$$L_1 = \frac{L_1 + 2}{2L_1 + 5} \implies L_1^2 + 2L_1 - 1 = 0 \implies L_1 = \sqrt{2} - 1 = L$$

(taking the positive root, since $a_k > 0$ forces $L_1 \geq 0$). The identical computation gives $L_2 = L$.

Step 6: Conclude. The odd subsequence $x_{2k-1} \rightarrow L$ and the even subsequence $x_{2k} \rightarrow L$. By Exercise 15 (if the even and odd subsequences of (x_n) converge to the same limit, then (x_n) itself converges to that limit), we conclude

$$x_n \rightarrow L = \sqrt{2} - 1.$$

■

$\{x_n\} \text{ converges, and } x_n \rightarrow \sqrt{2} - 1$

Q10. Limits involving n -th roots

Both parts below use the following lemma, which is *not* the Ratio Test or the Root Test (those are separate tests that decide whether a *series* $\sum a_n$ converges). This is a different, purely sequence-theoretic fact relating two limits attached to a positive sequence $\{a_n\}$: the limit of the ratios a_{n+1}/a_n and the limit of the n -th roots $a_n^{1/n}$. Since it isn't always given by name in course notes, here is a full proof so it can be quoted and justified on the exam.

Lemma. If $a_n > 0$ for all n and $\frac{a_{n+1}}{a_n} \rightarrow L$ for some $L > 0$, then $a_n^{1/n} \rightarrow L$.

Proof. Let $\varepsilon > 0$ with $\varepsilon < L$. Since $a_{n+1}/a_n \rightarrow L$, there is N such that

$$L - \varepsilon < \frac{a_{n+1}}{a_n} < L + \varepsilon \quad \text{for all } n \geq N.$$

Multiplying this chain of inequalities for $n = N, N+1, \dots, m-1$ (telescoping the product $a_m/a_N = \prod_{n=N}^{m-1} \frac{a_{n+1}}{a_n}$), for every $m > N$:

$$(L - \varepsilon)^{m-N} < \frac{a_m}{a_N} < (L + \varepsilon)^{m-N}.$$

So $a_N(L - \varepsilon)^{m-N} < a_m < a_N(L + \varepsilon)^{m-N}$. Taking m -th roots,

$$[a_N(L - \varepsilon)^{-N}]^{1/m}(L - \varepsilon) < a_m^{1/m} < [a_N(L + \varepsilon)^{-N}]^{1/m}(L + \varepsilon).$$

Let $C_1 = a_N(L - \varepsilon)^{-N}$ and $C_2 = a_N(L + \varepsilon)^{-N}$ — both fixed positive constants (independent of m). Since $C_1^{1/m} \rightarrow 1$ and $C_2^{1/m} \rightarrow 1$ as $m \rightarrow \infty$, taking $\limsup$ and $\liminf$ of the squeeze above gives

$$\limsup_{m \rightarrow \infty} a_m^{1/m} \leq L + \varepsilon, \quad \liminf_{m \rightarrow \infty} a_m^{1/m} \geq L - \varepsilon.$$

This holds for *every* $\varepsilon > 0$ (with $0 < \varepsilon < L$), so letting $\varepsilon \rightarrow 0$ forces

$$\limsup_{m \rightarrow \infty} a_m^{1/m} \leq L \leq \liminf_{m \rightarrow \infty} a_m^{1/m}.$$

Since $\liminf \leq \limsup$ always, both equal L , so $\lim_{m \rightarrow \infty} a_m^{1/m} = L$. ■

(a) $x_n = \sqrt[n]{\frac{(2n)!}{n!n!}}$

Solution. Let $a_n = \binom{2n}{n} = \frac{(2n)!}{n!n!}$, so $x_n = a_n^{1/n}$. Compute the ratio:

$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)!/[(n+1)!]^2}{(2n)!/(n!)^2} = \frac{(2n+1)(2n+2)}{(n+1)^2} = \frac{2(2n+1)}{n+1} \rightarrow 4.$$

By the Lemma above (with $L = 4$), $x_n = a_n^{1/n} \rightarrow 4$. ■

$x_n \rightarrow 4$

(b) $x_n = \frac{1}{n} \sqrt[n]{(2n+1)(2n+2)\cdots(2n+n)}$

Solution. Write

$$\ln x_n = -\ln n + \frac{1}{n} \sum_{k=1}^n \ln(2n+k) = \frac{1}{n} \sum_{k=1}^n \ln\left(2 + \frac{k}{n}\right).$$

This is a Riemann sum for $f(x) = \ln(2+x)$ on $[0, 1]$ with partition points $x_k = k/n$. As $n \rightarrow \infty$,

$$\ln x_n \rightarrow \int_0^1 \ln(2+x) dx.$$

Substitute $u = 2+x$:

$$\int_0^1 \ln(2+x) dx = \int_2^3 \ln u du = [u \ln u - u]_2^3 = (3 \ln 3 - 3) - (2 \ln 2 - 2) = 3 \ln 3 - 2 \ln 2 - 1.$$

Note $3 \ln 3 - 2 \ln 2 = \ln\left(\frac{3^3}{2^2}\right) = \ln\left(\frac{27}{4}\right)$, so

$$\ln x_n \rightarrow \ln\left(\frac{27}{4}\right) - 1.$$

Exponentiating,

$$x_n \rightarrow \frac{27}{4e}.$$

$x_n \rightarrow \frac{27}{4e}$

Q11. True/false: log limits

(a) $n \log\left(1 + \frac{1}{n+1}\right) \rightarrow 1$?

Solution. Write $t_n = \frac{1}{n+1} \rightarrow 0$. Use the standard limit $\lim_{t \rightarrow 0} \frac{\ln(1+t)}{t} = 1$:

$$n \ln\left(1 + \frac{1}{n+1}\right) = \frac{n}{n+1} \cdot \frac{\ln(1+t_n)}{t_n}.$$

As $n \rightarrow \infty$: $\frac{n}{n+1} \rightarrow 1$ and $\frac{\ln(1+t_n)}{t_n} \rightarrow 1$ (since $t_n \rightarrow 0$). By the algebra of limits, the product $\rightarrow 1 \cdot 1 = 1$. ■

(a) is TRUE (limit = 1)

(b) $n \log\left(1 + \frac{1}{n^2}\right) \rightarrow 1$?

Solution. Write $t_n = \frac{1}{n^2} \rightarrow 0$. As above,

$$n \ln\left(1 + \frac{1}{n^2}\right) = \frac{n}{n^2} \cdot \frac{\ln(1+t_n)}{t_n} = \frac{1}{n} \cdot \frac{\ln(1+t_n)}{t_n}.$$

Here $\frac{1}{n} \rightarrow 0$ and $\frac{\ln(1+t_n)}{t_n} \rightarrow 1$, so the product $\rightarrow 0 \cdot 1 = 0 \neq 1$. ■

(b) is FALSE (limit = 0, not 1)

Q12. Second-order linear recurrences (via geometric difference sequences)

(a) $u_{n+2} = \frac{2u_{n+1} + u_n}{3}$, $0 < u_1 < u_2$

Solution. Let $d_n = u_{n+1} - u_n$. Then

$$d_{n+1} = u_{n+2} - u_{n+1} = \frac{2u_{n+1} + u_n}{3} - u_{n+1} = \frac{u_n - u_{n+1}}{3} = -\frac{1}{3}d_n.$$

So $\{d_n\}$ is geometric with ratio $-\frac{1}{3}$: $d_n = d_1 \left(-\frac{1}{3}\right)^{n-1}$, where $d_1 = u_2 - u_1 > 0$.

Telescoping (summing a finite geometric series, as in Example 22),

$$u_n = u_1 + \sum_{k=1}^{n-1} d_k = u_1 + d_1 \sum_{k=1}^{n-1} \left(-\frac{1}{3}\right)^{k-1} = u_1 + d_1 \cdot \frac{1 - \left(-\frac{1}{3}\right)^{n-1}}{1 - \left(-\frac{1}{3}\right)} = u_1 + \frac{3d_1}{4} \left[1 - \left(-\frac{1}{3}\right)^{n-1}\right].$$

By Exercise 12(2) of the notes ($r^n \rightarrow 0$ for every $|r| < 1$, applied here with $r = -\frac{1}{3}$), $\left(-\frac{1}{3}\right)^{n-1} \rightarrow 0$, so by Theorem 9 (Algebra of Convergent Sequences),

$$u_n \rightarrow u_1 + \frac{3d_1}{4} = u_1 + \frac{3}{4}(u_2 - u_1) = \frac{4u_1 + 3u_2 - 3u_1}{4} = \frac{u_1 + 3u_2}{4}.$$

This closed form is an explicitly convergent expression, so it simultaneously proves $\{u_n\}$ converges and identifies its limit. ■

$$u_n \rightarrow \frac{u_1 + 3u_2}{4}$$

$$(b) \frac{2}{u_{n+2}} = \frac{1}{u_{n+1}} + \frac{1}{u_n}, \quad 0 < u_1 < u_2$$

Solution. First, $u_n > 0$ for all n by induction: $u_1, u_2 > 0$, and if $u_n, u_{n+1} > 0$ then $\frac{1}{u_{n+2}} = \frac{1}{2} \left(\frac{1}{u_{n+1}} + \frac{1}{u_n} \right) > 0$, so $u_{n+2} > 0$.

Set $v_n = \frac{1}{u_n}$ (well-defined since $u_n > 0$). The recurrence becomes

$$2v_{n+2} = v_{n+1} + v_n \implies v_{n+2} = \frac{v_{n+1} + v_n}{2}.$$

Let $e_n = v_{n+1} - v_n$. Then

$$e_{n+1} = v_{n+2} - v_{n+1} = \frac{v_{n+1} + v_n}{2} - v_{n+1} = \frac{v_n - v_{n+1}}{2} = -\frac{1}{2}e_n,$$

so $e_n = e_1 \left(-\frac{1}{2}\right)^{n-1}$ with $e_1 = v_2 - v_1 = \frac{1}{u_2} - \frac{1}{u_1}$.

Telescoping as before,

$$v_n = v_1 + e_1 \sum_{k=1}^{n-1} \left(-\frac{1}{2}\right)^{k-1} = v_1 + e_1 \cdot \frac{1 - \left(-\frac{1}{2}\right)^{n-1}}{1 - \left(-\frac{1}{2}\right)} = v_1 + \frac{2e_1}{3} \left[1 - \left(-\frac{1}{2}\right)^{n-1} \right].$$

By Exercise 12(2), $\left(-\frac{1}{2}\right)^{n-1} \rightarrow 0$, so by Theorem 9,

$$v_n \rightarrow v_1 + \frac{2e_1}{3} = v_1 + \frac{2}{3}(v_2 - v_1) = \frac{v_1 + 2v_2}{3} = \frac{1}{3} \left(\frac{1}{u_1} + \frac{2}{u_2} \right) =: v^* > 0.$$

Since $v_n \rightarrow v^* > 0$ and $u_n = 1/v_n$, Theorem 9 part (4) (Algebra of Convergent Sequences, reciprocal rule — applicable exactly because the limit $v^* \neq 0$) gives

$$u_n \rightarrow \frac{1}{v^*} = \frac{1}{\frac{1}{3} \left(\frac{1}{u_1} + \frac{2}{u_2} \right)} = \frac{3}{\frac{1}{u_1} + \frac{2}{u_2}}.$$

■

$$u_n \rightarrow \frac{3}{\frac{1}{u_1} + \frac{2}{u_2}}$$

Q13. $\sum x_n$ convergent, $x_n > 0$: four derived series

Tools used throughout: Theorem 3.10 (convergence $\implies$ terms $\rightarrow 0$); Theorem 3.16 (Comparison Test — stated for nonnegative series with $a_n \leq b_n$ for every n); Theorem 3.5 (adding or discarding finitely many terms doesn't affect convergence); Theorem 3.14 (a positive-term series converges iff its partial sums are bounded above).

(a) $\sum x_n^2$ converges

Solution. Since $\sum x_n$ converges, Theorem 3.10 gives $x_n \rightarrow 0$, so there is N with $0 < x_n < 1$ for all $n \geq N$. For such n , multiplying $x_n < 1$ by $x_n > 0$ gives $x_n^2 < x_n$.

Apply the Comparison Test (Theorem 3.16) to the tail series $\sum_{n \geq N} x_n^2$ and $\sum_{n \geq N} x_n$ (the hypothesis $a_n \leq b_n$ holds for every index once we start counting from N): since $\sum_{n \geq N} x_n$ converges (a tail of the convergent series $\sum x_n$, by Theorem 3.5), Theorem 3.16 gives $\sum_{n \geq N} x_n^2$ converges. By Theorem 3.5 again, adding back the finitely many terms $n < N$, the full series $\sum_{n=1}^{\infty} x_n^2$ converges. ■

(b) $\sum \frac{x_n}{n}$ **converges**

Solution. For every $n \geq 1$, $0 < \frac{x_n}{n} \leq x_n$ (since $\frac{1}{n} \leq 1$). This is precisely the hypothesis of the Comparison Test (Theorem 3.16), with $a_n = x_n/n$, $b_n = x_n$, holding for all n . Since $\sum x_n$ converges, Theorem 3.16 gives $\sum \frac{x_n}{n}$ converges. ■

(c) $\sum x_{2n}$ **converges**

Solution. Let $S_N = \sum_{n=1}^N x_{2n}$. Since every $x_k > 0$,

$$S_N = \sum_{n=1}^N x_{2n} \leq \sum_{k=1}^{2N} x_k \leq \sum_{k=1}^{\infty} x_k =: S < \infty,$$

the middle inequality holding because $\{x_2, x_4, \dots, x_{2N}\}$ is a sub-collection of the positive terms $\{x_1, x_2, \dots, x_{2N}\}$. So the partial sums (S_N) of $\sum x_{2n}$ are bounded above by S . By Theorem 3.14 (a positive-term series converges iff its partial sums are bounded above), $\sum x_{2n}$ converges. ■

(d) $\sum \frac{x_n}{1+x_n}$ **converges**

Solution. For every n , since $x_n > 0$ we have $1 + x_n > 1$, so

$$0 < \frac{x_n}{1+x_n} < x_n \quad \text{for all } n.$$

By the Comparison Test (Theorem 3.16), since $\sum x_n$ converges, $\sum \frac{x_n}{1+x_n}$ converges. ■

Q14. Convergence/divergence of ten series

Preliminary Lemma (Alternating Series), proved directly from the Cauchy Criterion in exactly the style of Example 3.13 of the notes — your uploaded notes introduce alternating series (Definition 3.7) but the excerpt provided does not state a named “Leibniz Test”, so we prove the fact outright rather than citing it as a black box.

Lemma. If (b_n) is a decreasing sequence of positive numbers with $b_n \rightarrow 0$, then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges.

Proof. Let s_n be the n -th partial sum. For $p \geq 1$,

$$s_{n+p} - s_n = (-1)^n [b_{n+1} - b_{n+2} + b_{n+3} - \dots \pm b_{n+p}].$$

Since (b_n) is decreasing, grouping the bracketed terms in consecutive pairs shows this alternating sum lies between 0 and b_{n+1} (exactly the computation done for $b_n = 1/n$ in Example 3.13), so

$$|s_{n+p} - s_n| \leq b_{n+1} \quad \text{for every } p \geq 1.$$

Given $\varepsilon > 0$, since $b_n \rightarrow 0$ there is N with $b_{N+1} < \varepsilon$. Then for all $n \geq N$ and all $p \geq 1$, $|s_{n+p} - s_n| \leq b_{n+1} \leq b_{N+1} < \varepsilon$. So (s_n) is a Cauchy sequence, and by the Cauchy Criterion (Theorem 3.8), the series converges. ■

Solution. (i) $\sum \frac{n^2 + 1}{(n+3)(n+4)}$. The general term $\frac{n^2 + 1}{n^2 + 7n + 12} \rightarrow 1 \neq 0$. By Theorem 3.10 (necessary condition: convergence $\Rightarrow$ terms $\rightarrow 0$), the series **diverges**.

(ii) $\sum \left[\sqrt[3]{n^3 + 1} - n \right]$. $\sqrt[3]{n^3 + 1} = n \left(1 + \frac{1}{n^3} \right)^{1/3} = n + \frac{1}{3n^2} + O(n^{-5})$, so the term $\sim \frac{1}{3n^2}$. By the Limit Form of the Comparison Test (Theorem 3.19) with $b_n = 1/n^2$: the ratio $\rightarrow \frac{1}{3} \in (0, \infty)$. Since $\sum 1/n^2$ converges (p -Series Test, Theorem 3.23, $p = 2 > 1$), so does the given series: it **converges**.

(iii) $\sum \frac{\log n}{2n^3 - 1}$. With $b_n = 1/n^2$: $\frac{\log n / (2n^3 - 1)}{1/n^2} = \frac{n^2 \log n}{2n^3 - 1} \rightarrow 0$. By Remark 3.22(1) (the 0-limit case of the Limit Comparison Test), since $\sum 1/n^2$ converges (Theorem 3.23), the series **converges**.

(iv) $\sum \sin^2 \frac{1}{n}$. Since $\frac{\sin(1/n)}{1/n} \rightarrow 1$ (standard trig limit), $\frac{\sin^2(1/n)}{1/n^2} = \left(\frac{\sin(1/n)}{1/n} \right)^2 \rightarrow 1$. By Theorem 3.19 (Limit Comparison Test) with $b_n = 1/n^2$: since $\sum 1/n^2$ converges (Theorem 3.23), the series **converges**.

(v) $\sum n \tan \frac{1}{n^2}$. Since $\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x} \rightarrow 1 \cdot 1 = 1$ as $x \rightarrow 0$, taking $x = 1/n^2$ gives $n^2 \tan \frac{1}{n^2} \rightarrow 1$, i.e. $\frac{n \tan(1/n^2)}{1/n} \rightarrow 1$. By Theorem 3.19 (Limit Comparison Test) with $b_n = 1/n$: since $\sum 1/n$ diverges (Theorem 3.23, $p = 1$), the series **diverges**.

(vi) $\sum \frac{(n!)^2}{(2n)!}$. By d'Alembert's Ratio Test (Theorem 3.24):

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(2n+1)(2n+2)} \rightarrow \frac{1}{4} < 1,$$

so the series **converges**.

(vii) $\sum \frac{(-1)^{n-1} n}{n^2 + 1}$. Let $b_n = \frac{n}{n^2 + 1}$. Then $b_n \rightarrow 0$, and b_n is decreasing for $n \geq 1$ since $f(x) = \frac{x}{x^2 + 1}$ has $f'(x) = \frac{1 - x^2}{(x^2 + 1)^2} \leq 0$ for $x \geq 1$. By the Lemma above, the series **converges**. It is not absolutely convergent: $\sum b_n = \sum \frac{n}{n^2 + 1}$ diverges by Theorem 3.19 (Limit Comparison with $\sum 1/n$, which diverges by Theorem 3.23). Hence the series **converges conditionally**.

(viii) $\sum \frac{(-1)^n 3^n}{n!}$. Absolute series $\sum \frac{3^n}{n!}$: by d'Alembert's Ratio Test (Theorem 3.24), $\frac{3^{n+1}/(n+1)!}{3^n/n!} = \frac{3}{n+1} \rightarrow 0 < 1$, so it converges. By Theorem 3.34 (absolute convergence $\Rightarrow$ convergence), the original series **converges absolutely**.

(ix) $\sum \frac{(-1)^{n-1} (n!)^2}{(2n)!} 5^n$. Let $u_n = \frac{(n!)^2 5^n}{(2n)!} > 0$. Then

$$\frac{u_{n+1}}{u_n} = \frac{5(n+1)^2}{(2n+1)(2n+2)} \rightarrow \frac{5}{4} > 1.$$

By Theorem 13 (this is the Ratio Test for *sequences*, not the series version): since $u_{n+1}/u_n \rightarrow \ell > 1$, we get $u_n \rightarrow +\infty$. In particular $u_n = |a_n| \not\rightarrow 0$, so $a_n \not\rightarrow 0$ either. By Theorem 3.10 (necessary condition for series convergence), the series **diverges**.

(x) $\sum \frac{(-1)^{n-1} \log(n+1)}{n+1}$. Let $b_n = \frac{\log(n+1)}{n+1}$. Writing $g(t) = \frac{\log t}{t}$ so $b_n = g(n+1)$: $g'(t) = \frac{1 - \log t}{t^2} < 0$ for $t > e$, so b_n is decreasing once $n+1 > e$, and $b_n \rightarrow 0$. By the Lemma above (applied from the point where b_n becomes decreasing; Theorem 3.5 confirms that finitely many initial terms don't affect convergence), the series **converges**. It is not absolute: for $n+1 \geq e$, $\log(n+1) \geq 1$, so $b_n \geq \frac{1}{n+1}$; since $\sum \frac{1}{n+1}$ diverges (a tail of the harmonic series, Theorem 3.23 together with Theorem 3.5), the Comparison Test (Theorem 3.16) gives $\sum b_n$ diverges. Hence the series **converges conditionally**. ■

Q15. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \log 2 \implies \sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = ?$

Solution. Partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. So

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} - \sum_{n=1}^{\infty} \frac{(-1)^n}{n+1}.$$

First sum: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = - \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = -\log 2$.

Second sum: let $m = n+1$ (so m runs from 2 to ∞ , and $(-1)^n = (-1)^{m-1} = (-1)^{m+1}$):

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n+1} = \sum_{m=2}^{\infty} \frac{(-1)^{m+1}}{m} = \left(\sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \right) - \frac{(-1)^2}{1} = \log 2 - 1.$$

Therefore

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = -\log 2 - (\log 2 - 1) = 1 - 2\log 2.$$

■

(A) $1 - 2\log 2$

Q16. $\sum_{k=0}^{\infty} \frac{k^2 + 3k + 1}{(k+2)!}$

Solution. Decompose the numerator to match the falling-factorial structure of $(k+2)! = (k+2)(k+1)k!$. Try

$$k^2 + 3k + 1 = A(k+2)(k+1) + B(k+2) + C.$$

Expanding: $A(k^2 + 3k + 2) + B(k+2) + C = Ak^2 + (3A+B)k + (2A+2B+C)$. Matching coefficients with $k^2 + 3k + 1$: $A = 1$, $3A + B = 3 \Rightarrow B = 0$, $2A + 2B + C = 1 \Rightarrow C = -1$. So

$$k^2 + 3k + 1 = (k+2)(k+1) - 1.$$

Hence

$$\frac{k^2 + 3k + 1}{(k+2)!} = \frac{(k+2)(k+1)}{(k+2)!} - \frac{1}{(k+2)!} = \frac{1}{k!} - \frac{1}{(k+2)!}.$$

So

$$\sum_{k=0}^{\infty} \frac{k^2 + 3k + 1}{(k+2)!} = \sum_{k=0}^{\infty} \frac{1}{k!} - \sum_{k=0}^{\infty} \frac{1}{(k+2)!} = e - \left(\sum_{m=0}^{\infty} \frac{1}{m!} - \frac{1}{0!} - \frac{1}{1!} \right) = e - (e - 2) = 2.$$

■

(C) 2

Q17. Given $xe^x = \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}$, **find** $\sum_{n=1}^{\infty} \frac{2^n n^2}{n!}$

Solution. The given fact is exactly $\sum_{n=1}^{\infty} \frac{nx^n}{n!} = xe^x$ (since $\frac{n}{n!} = \frac{1}{(n-1)!}$). Apply the operator $x \frac{d}{dx}$ to both sides:

$$x \frac{d}{dx} \sum_{n=1}^{\infty} \frac{nx^n}{n!} = \sum_{n=1}^{\infty} \frac{n \cdot n x^n}{n!} = \sum_{n=1}^{\infty} \frac{n^2 x^n}{n!},$$

and on the right,

$$x \frac{d}{dx} (xe^x) = x(e^x + xe^x) = xe^x(1+x).$$

So

$$\sum_{n=1}^{\infty} \frac{n^2 x^n}{n!} = x(1+x)e^x.$$

Setting $x = 2$:

$$\sum_{n=1}^{\infty} \frac{2^n n^2}{n!} = 2(1+2)e^2 = 6e^2.$$

■

(D) $6e^2$

Q18. Two more alternating series

Solution. $\sum \frac{(-1)^{n-1}}{n\sqrt{n}} = \sum \frac{(-1)^{n-1}}{n^{3/2}}$: The absolute series $\sum \frac{1}{n^{3/2}}$ is a p -series with $p = \frac{3}{2} > 1$, hence convergent. So the series **converges absolutely**.

$\sum \frac{(-1)^{n-1}n}{n^2+1}$: This is exactly Q14(vii): it **converges conditionally** (Leibniz Test gives convergence; the absolute series $\sum \frac{n}{n^2+1}$ diverges by Limit Comparison with the harmonic series). ■

Q19. Prove $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n(\log n)^{1/3}}$ **is conditionally convergent**

Solution. Let $b_n = \frac{1}{n(\log n)^{1/3}} > 0$ for $n \geq 2$.

Step 1 ($b_n \rightarrow 0$ and decreasing): As $n \rightarrow \infty$, $n(\log n)^{1/3} \rightarrow \infty$, so $b_n \rightarrow 0$. Also $f(x) = \frac{1}{x(\log x)^{1/3}}$ has both factors x and $(\log x)^{1/3}$ strictly increasing for $x > 1$, so f is strictly decreasing on $(1, \infty)$; hence b_n is decreasing.

By the **Leibniz (Alternating Series) Test**, $\sum_{n=2}^{\infty} (-1)^{n+1} b_n$ **converges**.

Step 2 ($\sum b_n$ diverges): Use the Integral Test with $f(x) = \frac{1}{x(\log x)^{1/3}}$, which is positive, continuous, and decreasing on $[2, \infty)$. Substituting $u = \log x$, $du = dx/x$:

$$\int_2^{\infty} \frac{dx}{x(\log x)^{1/3}} = \int_{\log 2}^{\infty} u^{-1/3} du = \left[\frac{u^{2/3}}{2/3} \right]_{\log 2}^{\infty} = \infty,$$

since the exponent $2/3 > 0$ makes the antiderivative diverge as $u \rightarrow \infty$. So the improper integral diverges, and by the Integral Test, $\sum b_n$ **diverges**.

Conclusion: the series converges (Step 1) but not absolutely (Step 2), so it is **conditionally convergent**. ■

Q20. Prove $\left(\frac{1}{2}\right)^2 - \left(\frac{1 \cdot 4}{2 \cdot 5}\right)^2 + \left(\frac{1 \cdot 4 \cdot 7}{2 \cdot 5 \cdot 8}\right)^2 - \dots$ **is conditionally convergent**

Solution. Let $c_n = \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{2 \cdot 5 \cdot 8 \cdots (3n-1)}$ and $a_n = c_n^2$, so the series is $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$.

Step 1 (a_n decreasing, $a_n \rightarrow 0$):

$$\frac{c_{n+1}}{c_n} = \frac{3n+1}{3n+2} < 1 \implies c_{n+1} < c_n \implies a_{n+1} < a_n,$$

so $\{a_n\}$ is strictly decreasing. For the limit, write $\frac{c_{n+1}}{c_n} = 1 - \frac{1}{3k+2}$ at step k , so

$$\ln c_n = \sum_{k=1}^{n-1} \ln \left(1 - \frac{1}{3k+2} \right) \leq - \sum_{k=1}^{n-1} \frac{1}{3k+2}$$

(using $\ln(1-t) \leq -t$ for $0 < t < 1$). Since $\sum_k \frac{1}{3k+2}$ diverges (compare with the harmonic series),

$\ln c_n \rightarrow -\infty$, so $c_n \rightarrow 0$ and hence $a_n = c_n^2 \rightarrow 0$.

By the **Leibniz Test**, the series **converges**.

Step 2 ($\sum a_n$ diverges, by Raabe's Test): The Ratio Test is inconclusive here since $\frac{a_{n+1}}{a_n} = \left(\frac{3n+1}{3n+2}\right)^2 \rightarrow 1$. Apply **Raabe's Test**: compute

$$\frac{a_{n+1}}{a_n} = \left(1 - \frac{1}{3n+2}\right)^2 = 1 - \frac{2}{3n+2} + \frac{1}{(3n+2)^2},$$

so

$$n \left(1 - \frac{a_{n+1}}{a_n} \right) = \frac{2n}{3n+2} - \frac{n}{(3n+2)^2} \longrightarrow \frac{2}{3} - 0 = \frac{2}{3}.$$

Since $\frac{2}{3} < 1$, Raabe's Test says $\sum a_n$ **diverges**.

Conclusion: the series converges (Step 1) but not absolutely (Step 2), so it is **conditionally convergent**. ■

Note on two ambiguous transcriptions from the scanned PDF: in **Q2(i)** the sequence was read as $\frac{3n}{3n+1}$ (to match the linear/linear style of Q2(ii)); in **Q7** the recursion for y_{n+1} was read as $\sqrt{x_n y_n}$ (the classical AGM problem). If your printed copy shows something different for either, flag it and I'll redo that part.